

PULSEGUARD TECHNOLOGIES INC. - STANDARD OPERATING PROCEDURE

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Title:            DATA BACKUP AND RETENTION POLICY
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Department:       Data & Analytics
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Approver:         Sneha Deshmukh (VP of Engineering / Data Committee)
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1. PURPOSE & OBJECTIVE

This Standard Operating Procedure (SOP) defines the mandatory backup schedules, backup encryption guidelines, and data retention rules for business records and production data at PulseGuard. The goal is to ensure high availability, guarantee data recovery capabilities in emergency scenarios, and strictly comply with financial and privacy retention laws.

2. SCOPE

This procedure governs all active databases, database archives, file storage buckets (S3), server snapshot backups, transaction logs, and customer chat histories owned or operated by PulseGuard Technologies Inc. It applies to all DevOps and DBA staff.

3. ROLES AND RESPONSIBILITIES

- 3.1. Infrastructure Engineer (DevOps)
 - Configures and monitors automated backup pipelines.
 - Verifies encryption key mappings in KMS.
 - Wipes expired customer records according to retention rules.
- 3.2. Database Administrator (DBA)
 - Tests backup integrity via weekly restoration trials.
 - Manages backup snapshot archiving tasks.
- 3.3. Chief Data Officer (CDO)
 - Review audit logs of backup completions monthly.
 - Authorizes exceptions to the standard retention schedules.

4. DEFINITIONS AND REFERENCES

- Incremental Backup: Capturing only the changes made since the last backup run.
- KMS (Key Management Service): A cloud service for managing encryption keys.
- AES-256: Advanced Encryption Standard with a 256-bit key length.
- SOP-DATA-001: Production Data Access and Privacy Protocols (reference).

5. MANDATORY BACKUP RULES AND FREQUENCY

Backup routines must be automated and operate on these schedules:

- 5.1. Daily Backup Schedule
 - Incremental backups must be executed every 24 hours at 1:00 AM UTC.
 - Failures in the daily backup pipeline must generate alerts to DevOps.
- 5.2. Monthly Backup Schedule
 - Full snapshot backups must be taken on the 1st of every month.
- 5.3. Backup Encryption Guidelines
 - All backup archives must be encrypted using AES-256.
 - Encryption keys must be stored in a separate KMS. Keeping keys on the same server as the backup archives is strictly forbidden.
- 5.4. Restoration Testing
 - DBA must perform a dry-run restoration test once every month.

6. DATA RETENTION PERIODS AND PURGING MANDATES
- Business records and logs must be stored and purged according to these rules:
- 6.1. Financial Logs Retention
 - All transaction logs, invoicing records, and tax logs must be retained for a minimum of 7 years.
 - 6.2. Customer Chat History Retention
 - Customer chat history logs must be retained for a maximum of 1 year.
 - Once a log exceeds the 1-year timeline, it must be purged automatically via scheduled DB cleanups.
 - Retention of chat history beyond 1 year is a compliance violation.
7. PROCEDURAL STEP-BY-STEP FLOW
- Step 1: Automated cron triggers daily backup script at 1:00 AM UTC.
 - Step 2: Script generates snapshot, compresses it, and encrypts via KMS AES-256.
 - Step 3: Encrypted backup is uploaded to a secure, write-once S3 storage bucket.
 - Step 4: Backup job sends completion status to the monitoring dashboard.
 - Step 5: On the 1st of the month, cron triggers the full monthly backup script.
 - Step 6: Weekly cleanup cron identifies chat logs older than 365 days.
 - Step 7: Cleanup cron runs DB purge command to permanently wipe expired chats.
 - Step 8: CDO signs off on monthly backup and retention verification report.
8. TROUBLESHOOTING & EMERGENCY RESTORATION
- 8.1. Backup Failure Response
 - If a daily backup fails twice in a row, the alert escalates to CDO.
 - 8.2. Escalation Contact Path
 - First Level: DevOps Alert System (devops-alerts@policypulse.dev).
 - Second Level: Chief Data Officer (Sanjay Mehta).
 - Executive Escalation: VP of Engineering (Sneha Deshmukh).
9. REVISION HISTORY LOG

Version	Date	Author	Description of Change
v1.0	2024-03-01	S. Mehta	Initial backup and retention SOP.
v2.0	2025-02-12	S. Mehta	Added AES-256 encryption requirements.
v2.4	2026-01-15	S. Mehta	Mandated 1-year chat history purges.